

INTRODUCTION:

I submit the following proposals to the Office of Science and Technology Policy regarding the Trump Administration's AI Action Plan to sustain and enhance American AI leadership. My suggestions fall under the following categories: *research initiatives*, *education*, *infrastructure*, and *healthcare*.

RESEARCH INITIATIVES:

Several federal agencies are already investing in cutting-edge AI research such as the NSF, DARPA, and NIST. These efforts are admirable and could be expanded to accelerate the development of generalizable knowledge in key areas such as healthcare or materials science.

These funding opportunities can incentivize researchers towards specific fields of study and in particular shape their approaches. To this end, we ought to consider what types of second-order effects may result from these efforts. For example, not requiring the labelling of synthetic content might accelerate research today but might have a secondary effect in that Americans may find it more difficult to distinguish “deepfakes” from authentic media in the future. Some tradeoffs are difficult to quantify yet should be considered before taking one side or the other. Moreover, there should be a structured and transparent decision-making approach when evaluating these tradeoffs and among reviewers. This can be achieved by completing the following five step plan:

1. Defining Tradeoff Criteria
 - a. Develop and explicitly list evaluation factors (cost, technical merit, risk, societal impact) and their relative weights in funding opportunity documents
 - b. Clearly state the relative importance of non-cost factors as “significantly more important”, “approximately equal to or “significantly less important” than cost
 - c. Use best practices from acquisition tradeoff processes as outlined in FAR.ⁱ
2. Enhance Reviewer Training and Guidance
 - a. Develop targeted training for review panel members on tradeoff analysis and Multiple Criteria Decision Analysis techniques.
 - b. Use case studies from prior reviews to illustrate effective tradeoff evaluations.
3. Integrate Structured Decision-Making Tools
 - a. Implement standardized forms and decision matrices within the review process to collect reviewers' assessments of each criterion.
 - b. Normalize and aggregate scores across multiple dimensions.
 - c. Require that the rationale for any decision to award a higher-priced proposal be clearly documented, as is done in FAR tradeoff processes.

4. Foster Collaborative Review Practices

- a. Host webinars or workshops for potential applicants and reviewers to clarify tradeoff evaluation criteria early in the funding cycle.
- b. Provide detailed feedback to applicants regarding tradeoff considerations in proposals.
- c. Establish a post-award review process to assess the impacts of the tradeoffs made during selection resulted in successful outcomes and use this data to refine criteria for future funding opportunities.

5. Enhance Transparency and Accountability

- a. Make evaluation criteria and tradeoff guidelines publicly available and easily accessible in funding opportunity announcements.
- b. Require that evaluation reports include a summary of how tradeoffs were assessed.
- c. Ensure that all tradeoff decisions are fully documented and reproducible.
- d. Implement an internal audit process to periodically review the consistency of tradeoff evaluations.

This plan provides a structured, transparent, and iterative approach to reviewing tradeoffs in AI funding opportunity design, drawing on practices from federal acquisition and evaluation frameworks. By using structured decision-making tools, federal agencies can enhance the overall quality of the funding process by better aligning resource allocation with national research priorities.

EDUCATION:

AI will have broad impacts across our society, and in some cases is already doing so, e.g. AI is increasing worker productivity. We need to equip our students and workforce with the skills and tools they need to succeed. Therefore, I suggest enacting the following plan to improve AI literacy and proficiency across every education level.

1. Approaches by Education Level:

- a. K-12:
 - i. Create a national framework (e.g., building on initiatives like AI4K12) that defines what AI literacy means at various grade levels, from basic algorithmic thinking in elementary school to ethical considerations in high school.ⁱⁱ
 - ii. Embed AI topics within math, science, social studies, and computer science curricula. For example, use data analysis projects in math classes.

- iii. Introduce hands-on projects and competitions (e.g., robotics challenges, coding clubs) to encourage experiential learning and problem-solving using AI/ML tools.
 - b. Higher Education:
 - i. Develop undergraduate and graduate courses that cover both the technical foundations (machine learning, data science) and societal implications (ethics, policy) of AI.
 - ii. Encourage student-led research or capstone projects that require designing and implementing AI-driven solutions, linking academic theory with real-world applications.
 - c. Lifelong Learning:
 - i. Offer short courses, boot camps, and online modules to help current professionals and non-traditional students update their AI skills.
- 2. Teacher Training and Professional Development
 - a. Establish nationwide workshops and seminars for teachers to deepen their understanding of AI concepts and modern educational tools.
 - b. Create centralized online repositories of AI teaching resources (curriculum guides, lesson plans, interactive simulations) that are regularly updated.
- 3. Infrastructure and Resources Development:
 - a. Invest in school-based AI labs and makerspaces equipped with computers, AI software, and robotics kits to facilitate hands-on learning.
 - b. Foster public-private partnerships to secure technology donations, guest lectures, and real-world project collaborations that expose students to current AI practices.
- 4. Quality Improvement:
 - a. Develop standardized assessments and project evaluations that measure both technical understanding and critical thinking about AI/ML.
 - b. Use surveys and performance metrics to gather feedback from teachers, students, and parents, refining curricula and teaching methods on a regular basis.
- 5. Implementation:
 - a. Establish a task force to define national AI literacy standards and develop a comprehensive framework as well as a suitable timeline for enactment.
 - b. Begin pilot programs in select districts, evaluate outcomes, and then scale successful initiatives nationwide.

By systematically integrating AI literacy into every educational level, we can empower students to thrive in an AI-driven future. This plan not only prepares students for emerging challenges but also positions our nation as a leader in AI education.

INFRASTRUCTURE:

The training and operation of AI models is resource intensive, and demand is rapidly increasing. The United States must create domestic conditions suitable for the continued development and operation of AI in order to sustain our advantage in this field. Therefore, this administration should plan to increase the supply of energy and AI infrastructure in the following ways.

1. Policy and Regulatory Reform:
 - a. Review and remove barriers from legacy policies that hinder rapid deployment of AI infrastructure, building on actions such as the “Removing Barriers to American Leadership in Artificial Intelligence” executive order.ⁱⁱⁱ
 - b. Collaborate with state, local, and tribal authorities to reduce or eliminate regulatory redundancies for vital infrastructure projects.
2. Infrastructure Modernization and Expansion:
 - a. Appropriate funds for or incentivize via tax credits the construction of new energy facilities, data centers, fiber-optic networks, and the modernization of existing electrical grid infrastructure.
3. Transparency and Stakeholder Engagement:
 - a. Publish progress reports and maintain an open data portal for stakeholders to track developments in AI infrastructure and energy supply projects

By integrating AI infrastructure development with strategic energy supply enhancements, the federal government can foster an ecosystem that supports U.S. technological leadership and economic growth.

HEALTHCARE:

AI has the potential to revolutionize healthcare, from improving diagnostic accuracy to optimizing the allocation of healthcare resources. However, its integration raises critical ethical concerns, including data privacy, bias, transparency, accountability, and informed consent. **Inadequate ethical safeguards can erode trust, which has been shown to correlate with a variety of negative outcomes,** such as decreased care quality, underutilization of healthcare resources, and critically, healthcare outcomes such as quality of life and patient satisfaction.^{ivv} The federal government must lead by establishing a robust regulatory framework that mandates ethical standards for AI use in healthcare. This

framework should integrate ethical principles from traditional sources as well as consensus-based formats such as FUTURE-AI.^{vi}

1. Objectives:
 - a. Ensure Patient Safety and Trust.
 - b. Protect Patients Rights and Privacy.
 - c. Establish Accountability and Transparency.
 - d. Foster Innovation with Ethical Oversight.
2. Policy Framework:
 - a. Develop and mandate a federal AI ethics framework that incorporates core ethical principles (beneficence, non-maleficence, autonomy, and justice) as well as novel criteria such as transparency, accountability, and explainability.
 - b. Establish an independent, multidisciplinary AI oversight board with representatives from all relevant stakeholder groups (healthcare providers, AI/ML experts, ethicists, patient advocates, and regulators) to oversee AI applications in healthcare and biomedical research.
 - c. Define liability clearly to ensure that responsibility is allocated properly among developers, providers, and institutions.
3. Data Governance:
 - a. Ensure that all AI systems used in healthcare comply with the relevant statutory requirements as defined by HIPAA and other privacy laws.
 - b. Mandate de-identification and anonymization protocols for patient data used in AI training and testing.
 - c. Create or empower an existing federal agency to oversee data governance practices for healthcare AI.
 - d. Establish guidelines for informed consent and data sharing to reduce risks of data misuse.
4. Transparency, Explainability and Oversight:
 - a. Require developers to document and share the design, training data, and performance metrics of AI systems and tools when they are to be used in providing patient care or biomedical research with human subjects.
 - b. Encourage the use of explainable AI (XAI) methods so that healthcare providers can understand and trust AI recommendations.^{vii}
 - c. Ensure that AI tools serve as decision-support systems, where final decisions are made by human providers.

- d. Design and implement training for healthcare professionals on interpreting AI outputs and their responsible use in healthcare.
- 5. Liability and Accountability:
 - a. Clearly define legal responsibilities and liability for harms resulting from the use of AI systems in healthcare. This includes establishing accountability for developers, manufacturers, healthcare institutions, and providers.
 - b. Impose meaningful penalties for non-compliance with ethical standards and data privacy laws.
- 6. Stakeholder Engagement:
 - a. Involve patients, providers, ethicists, and developers in the design and continuing review of AI ethical guidelines.
 - b. Organize workshops and deliberations to gather public input to ensure that policies reflect the needs and concerns of all stakeholders.
- 7. Education and Research:
 - a. Build on suggested proposals in the *Education* section, fund education and training programs for healthcare providers and AI developers on ethical AI use and design.
 - b. Support research on AI ethics, fostering academic partnerships and grants that promote best practices in the ethical design and application of AI in healthcare. See *Research Initiatives*.
- 8. International Collaboration:
 - a. Partner with international organizations such as the WHO and EU to align ethical standards and regulatory frameworks to facilitate sharing data across institutions.

This plan aims to ensure that AI technologies in healthcare are ethical and transparent, and by extension, trustworthy. By establishing governance structures, clear legal frameworks, data protection standards, and stakeholder engagement, the federal government can lead the way in harnessing AI's potential to improve patient care.

CONCLUSION:

In conclusion, the proposals outlined above form a strategy for sustaining American AI leadership across several domains. Through AI research initiatives, we can drive innovative discoveries and ensure that research conducted is aligned with our research priorities. Enhancing AI literacy and skills through systematic educational reforms will empower our current and future workforce to succeed in an AI-driven world. Furthermore, modernizing our national infrastructure and energy supply will provide the

foundation necessary to support advanced AI applications. Finally, by establishing a regulatory framework and oversight mechanisms in healthcare, we can protect patient and participant rights, mitigate bias, and ensure accountability in healthcare and biomedical research. Together, these proposals enshrine the ethical use of AI across critical sectors and position the United States to be a global leader in AI for years to come.

Signed,

Nicolas Jean-Louis Hargraves Oliver

FAIR USE STATEMENT:

“This document is approved for public dissemination. The document contains no business-proprietary or confidential information. Document contents may be reused by the government in developing the AI Action Plan and associated documents without attribution.”

Moreover, this document reflects only the opinion of the author and in no way reflects the opinion of his employer or institution(s).

CITATIONS:

- AI4K12. “AI4K12.” Accessed March 10, 2025. <https://ai4k12.org/>.
- Birkhäuser, Johanna, Jens Gaab, Joe Kossowsky, Sebastian Hasler, Peter Krummenacher, Christoph Werner, and Heike Gerger. “Trust in the Health Care Professional and Health Outcome: A Meta-Analysis.” *PLOS ONE* 12, no. 2 (February 7, 2017): e0170988. <https://doi.org/10.1371/journal.pone.0170988>.
- Crigger, Elliott, Karen Reinbold, Chelsea Hanson, Audiey Kao, Kathleen Blake, and Mira Irons. “Trustworthy Augmented Intelligence in Health Care.” *Journal of Medical Systems* 46, no. 2 (January 12, 2022): 12. <https://doi.org/10.1007/s10916-021-01790-z>.
- “FAR | Acquisition.GOV.” Accessed March 10, 2025. <https://www.acquisition.gov/browse/index/far>.
- LaVeist, Thomas A., Lydia A. Isaac, and Karen Patricia Williams. “Mistrust of Health Care Organizations Is Associated with Underutilization of Health Services.” *Health Services Research* 44, no. 6 (2009): 2093–2105. <https://doi.org/10.1111/j.1475-6773.2009.01017.x>.
- Lekadir, Karim, Aasa Feragen, Abdul Joseph Fofanah, Alejandro F. Frangi, Alena Buyx, Anais Emelie, Andrea Lara, et al. “FUTURE-AI: International Consensus Guideline for Trustworthy and Deployable Artificial Intelligence in Healthcare.” arXiv, July 8, 2024. <https://doi.org/10.48550/arXiv.2309.12325>.
- The White House. “Removing Barriers to American Leadership in Artificial Intelligence,” January 23, 2025. <https://www.whitehouse.gov/presidential-actions/2025/01/removing-barriers-to-american-leadership-in-artificial-intelligence/>.

ⁱ “FAR | Acquisition.GOV.”

ii “AI4K12.”

iii “Removing Barriers to American Leadership in Artificial Intelligence.”

iv LaVeist, Isaac, and Williams, “Mistrust of Health Care Organizations Is Associated with Underutilization of Health Services.”

v Birkhäuser et al., “Trust in the Health Care Professional and Health Outcome.”

vi Lekadir et al., “FUTURE-AI.”

vii Crigger et al., “Trustworthy Augmented Intelligence in Health Care.”